Megan Coyne

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Education

B.S. Computer Science and Mathematics, University of Dayton (Dayton, OH), May 2026

- Anticipated Magna Cum Laude, Honors with Distinction
- Concentration: Cyber Defense
- Cumulative GPA: 3.88
- Trustees’ Merit Scholarship, Dean’s List
- Pi Mu Epsilon (National Mathematics Honor Society)

Research Experience

Honors Thesis | University of Dayton Honors Program

Topic: *Constructing an Extreme Value Model to Predict Municipal Financial Risk Based on Corporate Bankruptcy*

Faculty Mentor: Dr. Gayan Warahena Liyanage, University of Dayton Mathematics Department

- Developed a statistical model using Extreme Value Theory (EVT) to assess municipal bankruptcy risk from corporate financial instability.
- Focused on Detroit as a case study, integrating economic data from Ford, GM, and Stellantis with municipal indicators.
- Applied the Generalized Extreme Value (GEV) distribution and Block Maxima method to capture tail risk in financial time series.
- Conducted cross-correlation and Granger causality analysis to explore predictive relationships between corporate distress and municipal decline.
- Built a generalizable modeling framework for cities economically more reliant on a single industry.
- Used Python libraries for time series modeling, statistical analysis, and data visualization.

Research Associate | Air Force Research Laboratory

Topic: *Visual Attention, Change Blindness, and Introductory Neural Networks*

Faculty Mentor: Dr. Kevin Scmidt, Air Force Research Laboratory

- Built a dynamic Python GUI to run image-based change blindness experiments and record participant input.
- Implemented randomized trial conditions, timer-controlled image transitions, and interactive user response capture.
- Designed and trained a basic neural network with HuggingFace and TensorFlow to analyze behavioral response data.
- Contributed to stimulus development, data logging protocols, and research exploring connections between attention, perception, and memory.

Work Experience

Software Engineering Intern | University of Dayton Research Institute — October 2022 – Present

- Collaborate with researchers at the Air Force Research Laboratory on classified projects involving algorithm development, data analysis, and intelligence modeling.
- Support CI/CD integration by configuring GitLab pipelines, authoring Dockerfiles, and containerizing software for demonstration and reproducibility.
- Contributed to “Project Tool,” a command-line utility designed to automate project scaffolding, dependency checks, and CI/CD configuration through intelligent templating.

Teaching Assistant | University of Dayton Mathematics Department August 2025 – Present

- Assist with instruction for Calculus III, including attending lectures, grading assignments, and hosting structured review sessions.
- Provide one-on-one tutoring and academic support in the university's Math Help Center, reinforcing core concepts and problem-solving techniques.

Enterprise Technology Intern | Ford Motor Company

May 2025 – August 2025:

- Developed a Go backend for an interactive app that visualizes Ford systems using annotated diagrams, with multiple levels of technical explanations.
- Built REST controllers to serve full data sets, specific markdown files, and associated images.
- Integrated a GitHub App to fetch documentation, then stored it in a GORM-structured database.
- Managed and debugged CI/CD pipelines during early-stage Golang adoption.
- Wrote unit tests in Go to ensure reliability and data accuracy.
- Presented the completed project to Ford’s Chief Architect using the app as the primary demo interface.

May, 2024 – August 2024:

- Designed a technical reference application to help over 1,000 engineers understand and adopt Ford’s internal coding conventions.
- Implemented backend functionality using Java Spring Boot; verified endpoint behavior through unit testing (JUnit5) and API tools (Postman).
- Supported deployment to Google Cloud Run and configured secure access via ADFS integration.
- Presented technical outcomes and project scope to engineering leadership.

Conferences

Nebraska Conference for Undergraduate Women in Mathematics, January 2024

- Selected as one of two students by the University of Dayton to represent the university.

Campus Involvement

Math Club | *President* August 2022 – Present

- Coordinate exciting lectures about math research and new discoveries, host math board game nights.
- Schedule the annual Math Department Picnic, organize the High School Math Competition, and assisted in planning the Putnam Competition on campus.

Sigma Kappa | *Vice President of Member Integrity* January 2023 – Present

- Social and service-based sorority taking part in activities to raise awareness for Alzheimer’s disease.
- Liaise between members and national leadership to support conflict resolution and uphold standards.

Flyer Consulting | *Consultant* January 2024 – Present

- Provide high quality, thorough business solutions to Dayton non-profits and small businesses.
- Communicate effectively with clients to recommend technical implementations for the mentoring program.

Technical Skills

Languages & Frameworks: Python, Java, Golang, C, C#, PHP, JavaScript, HTML/CSS, AngularJS
Tools & Platforms: Git, Docker, Postman, SQL, REST APIs, Google Cloud Run
Testing & Development: JUnit5, CI/CD Pipelines, Agile Methodologies
Machine Learning & Data: TensorFlow, HuggingFace, Scikit-learn; Neural Networks, Gradient Descent, Data Preprocessing, Tensors
Other: Microsoft Excel, LaTeX